MILITARY SPECIFICATION
COMPUTER SET, TACTICAL
AN/ASN-91B(V)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers the production phase requirements for a digital computer designated as Computer Set, Tactical AN/ASN-91(V).

1.2 Classification. The equipment covered by this specification AN/ASN-91B(V) Computer Set, shall consist of the following:

Type Designation	Applicable Paragraph	Name
CP-1208A(P)/ASN-91A(V)	3.5.1	Computer, Tactical 1/
C-7830A/ASN-91(V)	3.5.2	Control, Red Lights 2/
C-7831A/ASN-91(V)	3.5.2	Control, White Lights 2/
MT-4078/ASN-91(V)	3.5.3	Base Shock Mount, Electrical Equipment

1/ Herein referred to as the Computer
2/ Herein referred to as the Navigational Weapon Delivery Panel (NWDP) Each Computer Set includes one type NWDP, or the other, not both.

1.3 Associated equipment. This equipment shall be capable of operating with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS.

2.1 Government documents. The following documents of the exact issue shown form a part of this specification to the extent specified herein. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall have precedence.

Comments or recommendations which may be of use in improving this document may be sent to the preparing activity when a Self-Addressed Standardization Document Improvement Proposal (DD Form 1426) appears at the end of this document, or by letter addressed to the cognizant activity.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Military

MIL-P-116F	Preservation - Packaging, Methods of
MIL-W-5088B	Wiring, Aircraft, Installation of
MIL-E-5400F, Notice 1	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422E	Testing, Environmental Aircraft Electronic Equipment
MIL-M-7793C, Notice 1	Meter, Time Totalizing
MIL-I-8500B	Interchangeability and Replaceability of Component Parts for Aircraft, Installation of
MIL-D-8706	Data and Test Engineering, Contract Requirements for Aircraft Weapons Systems
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303A	Test Procedures; Preproduction and Acceptance, for Aircraft Electronic Equipment, Format for
MIL-N-18307C(ASG)-3	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment
MIL-C-26482C(NAVY)-1 Supplement 1	Connectors, Electric, Circular, Miniature, Quick Disconnect
MIL-C-26518	Connectors, Electrical, Miniature, Rack and Panel, Environment Resisting, 200°C Ambient Temperature
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-D-1000	Drawings, Engineering and Associated Lists

STANDARDS

MIL-STD-130B	Identification Marking of U.S. Military Property
MIL-STD-198	Capacitors, Selection and Use of
MIL-STD-199	Resistors, Selection and Use of
MIL-STD-210A	Climate Extremes for Military Equipment
MIL-STD-461A, Notice 3	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic, Interference Characteristics, Measurement of
MIL-STD-681	Identification Coding and Application of Hook Up and Lead Wire
MIL-STD-701	List of Standard Semiconductor Devices
MIL-STD-704A	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-756	Reliability Evaluation from Demonstration Data
MIL-STD-781B	Reliability Tests Exponential Distribution
MIL-STD-794	Part and Equipment, Procedure for Packing and Packaging
MIL-STD-882	System Safety Program for Systems and Associated Subsystem and Equipment, Requirements for
MIL-STD-1472A	Human Engineering Design Criteria for Military Systems Equipment and Facilities
MS-17322B	Meter, Time Totalizing, 115 Volt, 400 Cycle

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

ANA(MIL) Bulletin-400 Airborne Electronic and Associated Equipment, Applicable Documents.

2.2 Nongovernment documents. The following documents of the exact issue shown shall form a part of the specification to the extent specified herein. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall have precedence.

SPECIFICATIONS

IBM

6009704	Microcircuits, Monolithic and Hybrid, Procurement for
6009484	Printed Wiring Boards, Plated through Hole, Specification for
6850012	Multilayer Interconnection Board (MIB), Requirements for
6870210	Plate, Red Lighting Panel, Requirements for
6870299	Plate, Red Lighting Panel, Requirements for
6877508	Computer, TC2A, Nav-Weapon Delivery, Signal Interface for A7 D/E Airplane.

DRAWINGS

IBM

6877502	Digital Computer (TC-2A) Installation Drawing
6877727	Installation Drawing, NWDP, Modified

OTHER PUBLICATIONS

IBM Procedures

76-329-2408	TC-2A Burn In Procedure
76-329-2409	TC-2A Pre-Burn In Procedure
76-329-2410	TC-2A Navigation/Weapon Delivery Computer Acceptance Test Procedure
76-329-2413	Volume II TC-2A TAAF Test Procedure
76-329-2440	TC-2A Factory Acceptance Test Procedure

2.3 Availability of documents. When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the following locations.

GOVERNMENT DOCUMENTS

Commanding Officer, Naval Supply Dept, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

IBM DOCUMENTS

International Business Machines, Electronic Systems Center, Owego, New York 13827

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing (Environmental Qualification) and reliability growth testing (TAAF).

3.2 Parts and materials. The following objectives shall be given prime consideration in the design of the Computer Set.

a. Microelectronic integrated circuits (MIC) shall be used as a prime functional element of the equipment. In circuits where state-of-the-art does not permit the use of MIC, the design shall be the latest conventional solid-state form that will lend itself to replacement of MIC units on a circuit, card, or module basis, when available. Circuits shall be designed to use a minimum of different parts, subassemblies, cards, and modules.

b. The input/output section shall use the solid-state circuits defined in IBM Specification 6877508. Circuits with improved performance characteristics, including compatibility with the equipment listed in paragraph 6.6, may be used when approved by the cognizant Navy office. No electromechanical conversion devices shall be utilized, except for the 28 volt DC discrete outputs and the series relay in the Synchro outputs to the Horizontal Situation Indicator (HSI). The coils on electromagnetic components when used shall be suppressed or isolated from the aircraft interface. The wiring associated with each input/output circuit shall be sized so each wire will be protected by the aircraft circuit breakers or the component(s) in the Computer Set circuits, e.g., when the output from a relay contact which is connected to +28VDC is shorted to ground, the wiring in the Computer Set shall not fail.

c. Other parts and materials requirements shall conform to MIL-E-5400.

d. Nonrepairable subassemblies, as outlined in MIL-E-5400, shall only be used when practicable. The general size of the subassembly and the amount of circuitry to be included shall be approved by the cognizant Navy office. Nonrepairable subassemblies must be reliable (See 6.7).

3.2.1 Parts-Material selection (new designs). Parts shall be selected using the following, in order of precedence:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the currentversion before use.

a. ANA Bulletin 400
(1) MIL-STD-701, 198 and 199
(2) Other military specifications

b. Parts common with ASN-91(V) Computer Set

c. Parts used on other current avionics and missile programs (i.e., F-15, F-14, F-111, EA-6B, A7, A6, Harpoon Missile, etc.).

d. New Nonstandard Parts.

The Contractor recognizes the benefits to be derived by an aggressive part standardization program and will use the ASN-91(V) Computer Set parts list extensively during the selection process.

JAN TX and ER specifications will be utilized wherever possible and integrated circuits shall be screened to IBM Specification 6009404 as on the ASN-91(V) Computer Set.

3.2.2 Nonstandard part approval. Nonstandard parts approved on current avionics and missile programs (F-14, F-15, A7, F-111, A6, EA-6B, Harpoon, etc.) are considered approved for use on this computer without further action. These parts are currently in or will be added to government inventory.

3.3 Design and Construction. The Computer Set shall conform with all the applicable requirements of MIL-E-5400 for design, construction, and workmanship except as otherwise specified herein.

3.3.1 Total weight. The total weight of the Computer Set, excluding external cables, shall be a minimum and shall not exceed 89.4 pounds with 64K words of storage.

3.3.2 Reliability.

3.3.2.1 Operational stability. The Computer Set shall operate with satisfactory performance, continuously or intermittently, for a period of at least 500 operational hours or 6 months, whichever occurs first, without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating life. The Computer Set shall have a minimum total operating life of 4000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the Contractor and subject to approval by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

3.3.2.3 Storage life. The Computer Set shall meet the performance and service life requirements of the specification after five years storage under world-wide climate conditions specified in MIL-STD-210 when packaged in accordance with MIL-STD-794, Level A and MIL-P-116, Method IIA.

3.3.2.4 Reliability in mean time between failure (MTBF). The Computer, with 32K 16-bit memory, shall have a MTBF of 750 hours (1100 hours with 200 hours burn in) and the NWDP shall have a MTBF of 9000 hours (with 200 hours burn in) when operating in the following operational profile and after being tested and accepted as outlined under the requirements of 4.4 The MTBF is based on an operational profile consisting of 75 flight hours per month, 2 ground operational hours per flight hour and 495 off hours per month. The MTBF value as specified shall be attainable after exposure to the specified storage conditions and shall not be degraded below the required level as a result of acceptable field usage and maintenance. The component part failure rates used in determining the above MTBF shall be obtained from the IBM Failure Rate Data Bank. In cases where neither observed data nor reasonable predicted rates are available, failure rates shall be synthesized, based upon a summation of the failure rates of component part elements.

3.5.2.5 Reliability growth through TAAF test. TAAF Reliability Growth Tests shall be performed in accordance with an approved TAAF Reliability Growth Phase Test Plan prepared in accordance with MIL-D-8706 and IBM Test Procedure 76-329-2413 prepared using MIL-STD-781 as a guide. TAAF test requirements shall be in accordance with 4.5.

3.3.2.5.1 Reliability preduction and analyses. An analytical reliability prediction shall be prepared and submitted which will provide a quantitative assessment of the equipment reliability. The prediction will be made on a paper design of the test sample in accordance with the design prediction procedure of MIL-STD-756. The prediction will provide a value reliability (MTBF, probability of mission success, etc.) that is greater than the specified value of reliability. The sources of failure rate data and prediction techniques shall be identified.

3.3.2.5.2 Thermal survey. A thermal survey shall be performed in accordance with MIL-STD-781 prior to the start of the TAAF Reliability Growth Test.

3.3.2.5.3 Test readiness review, status review, and reports. A Test Readiness Review shall be conducted at least seven (7) days prior to the start of TAAF Reliability Growth Phase testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

Monthly Status Reviews shall be held through the duration of the actual TAAF Reliability Growth Test to review test status with regards to progress, growth assessments, potential problems, corrective actions and review and assignment of action items.

Monthly Status Reports shall be submitted outlining the test results and all pertinent data obtained during that reporting period.

Failure reports shall be submitted on occasion of each failure containing the necessary information to permit determination of the origin and possible correction of failures.

A final report shall be submitted which sumarizes all results obtained during the TAAF Reliability Growth Testing. The report shall include a comparison of results with the reliability predictions, the conclusions drawn, a summary of problems and failures and remedial, preventive and corrective actions taken, and an assessment of the reliability achieved.

3.3.2.6 Time totalizing meter. The Computer shall contain an elapsed time totalizing meter which meets the requirements of MS17322 in accordance with MIL-M-7793 except as modified with respect to mounting, size, and configuration. The meter shall be located such that it may be read without the removal of a cover or any other part from the unit. The meter shall indicate "ON" time which is any operating mode including standby.

3.3.3 Cabling and connections. The Computer Set shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.1 Interconnection cabling. The Computer Set shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088 and IBM Specification 6877508. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.3.2 Connectors. Connectors for joining of external aircraft wiring to the Computer Set shall in accordance with MIL-C-26482 and/or MIL-C-26518 as applicable.

3.3.3.3 Internal wiring identification coding. Identification coding shall be used for all chassis wiring and shall be in accordance with MIL-STD-681, except that signal-carrying miscellaneous wires shall be in solid color. Safety ground shall be in solid green.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

3.3.4 Control panel. The NWDP shall conform to the applicable requirements of IBM Specification 6870210 and/or 6870299.

3.3.5 Interchangeability. In addition to the requirements of MIL-E-5400, all contractor parts, subassemblies, and WRA having the same part number shall be interchangeable with each other with respect to installation and performance. Changes in contractor's part number shall be governed by the drawing number requirements of MIL-D-1000. Parts interchangeability shall be in accordance with MIL-I-8500.

3.3.6 Electromagnetic interference. The Computer Set set shall meet the requirements of MIL-STD-461 with the following exception.

a. CS06 Conducted Susceptibility, Spike, Power Leads

3.3.7 Maintainability. The design, configuration, installation, and operation of the Computer Set shall minimize servicing requirements, including the need for special tools and support equipment, personnel skills, and manpower. Design shall permit maintenance at organization and field maintenance levels. Scheduled maintenance shall be held to a minimum. The use of encapsulated or nonrepairable modules will be held to a minimum.

3.3.7.1 Maintainability requirements. Organizational level maintenance, using approved co-resident Operational Test Program (OTP) and excluding access and close up (function of aircraft installation), shall not exceed 0.4 direct man-hours mean-time-to-repair (MTTR). Field (intermediate) level maintenance shall not exceed 2.75 man-hours mean-time-to-repair (MTTR). No more than 4.0 man-hours of direct time measured at the 95th percentile shall be required to repair any single failure. Preventive maintenance shall not be required. Repair times were computed on the basis of accomplishment by maintenance personnel equivalent to Air Force skill level "five (5)" or Navy "A" school graduates. All organizational level maintenance must be capable of being performed under severe operational environments using only standard hand tools.

3.3.7.1.1 Repair time. Direct man-hours repair time shall be defined as the sum of the Active Repair Time specified below:

a. Preparation time

b. Fault location time to failed SRA/WRA

c. Remove/replace time associated with failed SRA/WRA

d. Adjustment/calibration time when required

e. Final test time

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

Item procurement time and any sealant cure time associated with repair shall be specifically excluded.

3.3.7.2 Human performance/human engineering. The requirements of MIL-E-5400 shall apply. MIL-STD-1472 and MIL-H-46855 shall be utilized as guidelines in applying human engineering design criteria.

3.3.7.2.1 Special tools. In addition to the requirements of MIL-E-5400, where practicability requires the use of special tools, such tools shall be available from the contractor. Any special tool requirements shall be specified by the contractor and approved by the cognizant Navy office.

3.3.7.2.2 Handling equipment. The Computer shall be provided with integral external handles to permit transport to and from the aircraft with no requirement for a handling fixture.

3.3.7.3 Maintenance concepts. Design shall be such that maintenance can be performed under the following concepts:

3.3.7.3.1 Scheduled maintenance. Consists of preflight, postflight, periodic phase and special inspections. Scheduled maintenance shall be performed at organizational maintenance level.

3.3.7.3.2 Unscheduled maintenance.

3.3.7.3.2.1 Organizational maintenance level (on aircraft). Consists of verifying and isolating a fault in the Computer Set, removing and installing a like serviceable item and performing operational checks to verify proper system operation.

3.3.7.3.2.2 Field maintenance level (off aircraft). Consists of bench checks, disassembly, and repair of unserviceable WRA by replacement of discrepant subassemblies. Serviceability of all repaired items will be verified prior to return to serviceable stocks.

3.3.7.4 Automatic checkout equipment The Computer Set shall be compatible with Automatic Checkout Equipment when used in conjunction with suitable interfacing equipment and shall include test provisions to permit diagnostic and performance verification testing at intermediate and depot level maintenance. Subsequent reinstallation of each WRA into the system shall not require system alignment or adjustment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

3.3.7.4.1 External test points Each WRA shall have sufficient accessible external test points that, when used in conjunction with the operating connector, will: (1) Permit quantitative determination of the performance level of the WRA, and (2) if substandard performance is determined, permit isolation of fault to the removable subassembly. Subassembly refers to the first level of mechanical disassembly (pluggable assembly, module, etc.) within the WRA.

3.3.7.4.2 Internal test points. Each level of disassembly within the WRA shall have exposed or accessible internal test points when disassembled to the extent necessary for intermediate and depot level maintenance. These internal test points shall be sufficient to fulfill the requirements of External Test Points (above) except for the level below the lowest repairable assembly. That is, a packaging unit (module, board, PC card, etc.) that is designated Not Repairable, need not meet the requirements for fault isolation.

3.3.7.5 Self-test requirements. The Computer Set shall contain built-in self-test capabilities to permit automatic fault detection and isolation to the WRA. A mechanical fault indicator shall be provided for each WRA. Once set under program control, the indicator shall be visible after removal of power. A malfunction in any WRA shall also cause an indicator light to be illuminated on the advisory panel. The self-test feature shall be fail-safe, and the fault indicators shall be energized in the event of a malfunction of the self-test circuitry. The mechanical fault indicator for both the Computer and the NWDP shall be located on the Computer.

3.3.7.5.1 Manually initiated self-test. A capability shall be provided to place the Computer in a self-test mode by operation of an external control. In this mode, the Computer will not be required to perform other normal functions. In self-test mode, the Computer shall be capable of being programmed to provide detection of 90 percent of all Computer failures.

3.3.7.5.2 In-Flight self-test. The Computer shall be capable of being programmed to perform periodically, without interrupt of other functions, monitoring and test routines automatically during normal computer operations. Maximum practical fault detection shall be provided. The Computer shall continue to operate and process incoming data when a failure is detected which does not affect program operation. It shall automatically resume normal operation and provide outputs upon clearance of the failure.

3.3.7.5.3 BITE features supporting self-test. Software controlled BITE features shall be provided in the hardware to allow testing and control

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

under test of the I/O channels, program accessible clocks and counters, memory protection, and other hardware items as applicable. The BITE features shall be sufficient to allow complete periodic maintenance at the organization level for isolating faults to the WRA while installed in the aircraft when the appropriate software is loaded in the Computer.

3.3.8 Identification of product. Nameplate approval, identification marking, serial number assignment and nomenclature for the Computer Set Units shall be in accordance with MIL-N-18307. Items not requiring nameplates shall be marked for identification in accordance with MIL-STD-130.

3.3.9 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ±5°C)
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Reference Signal	26 VAC + 2% - 5%, Phase C, 400 Hz
Input Power Voltage	115 VAC + 2% - 5%, 3Ø, 400 Hz 27.5 ± 0.5 VDC

3.3.10 Service conditions. The Computer Set shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for the class of equipment defined herein and except as modified in paragraphs 3.3.10.1 through 3.3.10.4.

Equipment	Class	Category	Operation
Computer, Tactical	2X	III	Continuous
Control Panel	1	III	Continuous

3.3.10.1 Vibration.

3.3.10.1.1 Computer. Curve I of Figure 5, MIL-E-5400, is extended to 2000 Hz at the ±10g level for vibration, when installed on the Base Shock Mount.

3.3.10.1.2 NWDP. The following spectrum of sinusoidal vibration levels shall be the requirements for the NWDP:

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:48Z
Check the source to verify that this is the current version before use.

Frequency (Hz)	Displacement (Double Amplitude)	Acceleration (Peak g's)
10 to 14	0.2 inch	---
14 to 33	---	2g
33 to 60	0.036 inch	---
60 to 100	---	5g
100 to 120	0.098 inch	---
120 to 200	---	7g
200 to 240	0.0034 inch	---
240 to 2000	---	10g

3.3.10.2 Cooling air provisions. The Computer will be supplied conditioned air through a shroud, around but not connected to the Computer air inlet. Fan(s) shall be provided by the contractor, integral with the Computer, to circulate the conditioned air through the Computer. With the air conditioner inoperative, compartment ambient air would be circulated through the Computer by the fan(s) and the Computer shall continue to operate for a minimum of 30 minutes under these conditions in a degraded mode at MIL-E-5400, Class I temperature environment. The normal compartment ambient air in flight when cooling air is supplied is a MIL-E-5400, Class 2, environment. The degraded mode will not affect the digital portions of the Computer but may reduce the accuracy of the analog input/output channels by a factor of 2. Conditioned air supplied through the shroud during normal flight operation shall be as follows:

Altitude (Feet)	Ambient Pressure (PSIA)	Supply Air Temperature Dry Bulb (°F)	Supply Air Temperature Dry Air Rated (°F)	Supply Air Flow lb/min
Sea Level	14.696	45	32	1.15
5,000	12.228	42	32	1.05
8,000	10.916	40	32	0.99
10,000	10.106	39	32	1.12
15,000	8.294	36	32	1.23
20,000	6.753	32	32	1.24
22,500	6.075	32	32	1.24
25,000	5.454	30	30	1.20
30,000	4.364	18	18	1.10
35,000	3.458	0	0	1.04
40,000	2.720	-14	-14	0.98
45,000	2.139	-13	-13	0.92
50,000	1.682	-12	-12	0.87

3.3.10.2.1 Ground operation. Ground operation of the Computer without conditioned cooling air should not exceed 30 minute intervals when inlet air temperatures are in excess of 131°F. Exceptions to these conditions will degrade the MTBF of the unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

3.3.10.2.2 Thermal overload protection. The Computer shall shut down automatically (by opening the 28 VDC circuit to the external power contactor) to prevent damage due to internal temperature excursions beyond safe operating limits. The Computer shall resume normal operation when temperature conditions return to safe operating level.

3.3.10.3 Altitude. The Computer Set shall be capable of operating up to a maximum of 50,000 feet.

3.3.10.4 Shock. NWDP shock requirements shall be as follows:

Peak Acceleration	Time to Peak Acceleration
15g	0.0025 sec
21g	0.00125 sec
27g	0.00083 sec
31.5g	0.000675 sec
45g	0.0005 sec
60g	0.0004 sec

These shock requirements do not supersede the requirements in MIL-E-5400 but are additional requirements.

3.3.11 Warm-up time. The time required for the Computer to warm up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under the aircraft installed environment as specified herein.

3.3.12 Input power requirements. The Computer Set shall meet all applicable requirements of MIL-STD-704 for category B equipment and shall provide specified performance from power sources with characteristics as defined in MIL-STD-704. The required power contactor to control the input power will be mounted in the aircraft with control signals through the NWDP and through the Computer. The effects of voltage transients in accordance with MIL-STD-704 on equipment performance shall be as defined in 3.5.1.8.1 and 3.5.1.14. Any degradation of performance beyond that specified must be approved by the cognizant Navy office.

The power required shall not exceed the following amounts:

a. AC Power: 115 VAC, 3 Phase, 400 Hz, Category B
580 VA at .76 P F (441 watts 64 K memory)

b. DC Power: 28 VDC Category B. Up to 7.0A (all dissipated in external loads).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

3.3.12.1 Under voltage protection. The Computer Set shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3.13 Multilayer interconnection boards and PC boards. All new MIBs and PC boards shall meet the requirements of IBM Specification 6009484. Existing MIBs and PC boards shall meet the requirements of IBM Specification 6850012 or 6009484 as specified on the applicable drawing.

3.3.14 Explosive atmosphere. The Computer Set shall be designed to operate in the presence of an explosive atmosphere without creating an explosion or to contain an explosion occurring inside the equipment. Since equipments operate in ever changing potentially explosive atmospheres, the equipments, when being laboratory tested, must operate in the presence of the optimum fuel-air mixture which requires the least amount of energy for ignition. The equipment igniting energy may be produced electrically, thermally or chemically. Components such as relays, switches, and motors with commutators, which in normal operation produce, or are likely to produce, sparking or arcing shall either be of a sealed design or placed in sealed containers.

3.3.15 Hazards. All potential hazards shall be reduced to a minimum. The action to eliminate or control hazards shall be in accordance with MIL-STD-882, Para. 5.9. Hazard levels are defined in MIL-STD-882, Para. 3.14. An explanation shall be presented to the contracting officer for any hazards that cannot be eliminated or controlled and all hazard levels except Category I shall require the approval of the contracting officer.

3.4 Performance. Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under standard conditions will be specified.

3.4.1 Instruction set. The basic instruction set shall include, as a minimum, all of the instructions in Table I. The instruction execution times in Table I shall be the maximum allowable for an instruction not including interference, such as interrupts or I/O cycle steals.

3.4.1.1 Input/output control. The instruction set shall include an Input/ Output Control (IOC) instruction and a Control Input/Output (CIOS) instruction which will control the flow of data to (Write) and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

from (Read) the associated equipment and NWDP. An exception is I/O that is associated with Direct Memory Access (DMA). See 3.5.1.12 and 3.5.1.13 for descriptions.

3.4.2 Accuracy. The data word length shall be defined for each of the digital I/O interfaces in 3.5.1.12.

The A/D and D/A conversion accuracies and the number of bits converted shall be as defined in 3.5.1.13.

The data word length within the processor section shall be as defined in 3.5.1.6.

3.4.3 Interface signal characteristics. The interface signal characteristics shall be identical to the CP-952/ASN-91V Computer, where applicable. The interface signal characteristics shall be included and documented in IBM Specification 6877508.

3.5 Detail requirements.

3.5.1 Computer, tactical. The computer shall meet the following requirements.

3.5.1.1 Function. The Computer, when connected to the associated equipment and the NWDP, shall receive analog and digital data from those units, process the data in real time, thereby performing the functions listed in 3.4.1 and output the necessary data in the form suitable for commands, control, and indications to the associated equipments.

3.5.1.1.1 CP-952/ASN-91(V) program compatibility. The Computer, when installed in an A-7D/E aircraft, shall perform all the functions of the CP-952/ASN-91(V) with minimal modification to the flight program used in the CP-952/ASN-91(V) when installed in the Computer. No aircraft wiring changes shall be required for the application of the Computer in place of a CP-952/ASN-91(V) Computer.

3.5.1.2 Form factor. The Computer shall conform to the dimensions specified in IBM Drawing 6877502.

3.5.1.3 Weight. The Computer shall not exceed 74 pounds with 65,536 words of storage.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

3.5.1.4 Contents. The Computer shall contain the following functional sections:

a. Processor

b. Memory

c. Digital I/O

d. Analog I/O

e. Power Supply

3.5.1.5 Connections. The Connections to/from the Computer shall be as documented in IBM Specification 6877508.

3.5.1.6 Organization. The Computer shall be a stored program, fixed point, general purpose computer. Data representation shall be two's complement. The operand word length shall be 16 bits (15 plus sign) for single precision arithmetic and 32 bits (31 plus sign) for double precision arithmetic.

3.5.1.7 Computational speed. As a minimum, the Computer shall be capable of executing 454,100 operations per second. The speed shall be calculated using the computer mean execution times for single precision, direct address without interference, with an instruction mix of 75% ADD (1/2 Indirect), 18% BRANCH (1/2 Taken), 5% MPY (1/2 Indirect) and 1% DIVIDE (1/2 Indirect).

3.5.1.8 Memory. The type of memory shall be magnetic core storage. The storage module shall contain 16,384 words (16 bits of data plus 1 parity bit) of random access memory. The storage shall be expandable in 16,384 word modules to 65,536 words. A memory protection feature shall be incorporated which shall prohibit alteration of all but the 0-4095, 16384-20479, 32768- 36863, and 49152-53247 addresses of memory. The lower 1024 locations of each 16,384 word section shall be directly addressable and shall be selected by the location of the executing program. Each directly addressable sector shall contain, as the lower eight locations, program linkage and base registers which are the active registers when they reside in the active directly addressable sector. They shall be selectable by status register control.

3.5.1.8.1 Data protection. The Computer shall contain the necessary circuitry to protect memory data from any normal or abnormal input power

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

transients as defined in MIL-STD-704. A shutdown procedure that will protect the data within the Computer's memory shall be initiated when abnormal power transients are detected. When normal power is restored, a restart sequence shall cause the Computer to return to a start-up point in the program. Initialization will be effected as required. Data from external sources that would have arrived during the power interruption will be lost. The turn-on sequence shall not require more than 100 milliseconds.

3.5.1.8.1.1 28 VDC transients. A voltage transient of .050 seconds, as defined by limit 3 of Figure 9 in MIL-STD-704 on the 28-volt discrete inputs to the Computer, will be considered an abnormal power condition under which the Computer is not required to operate.

3.5.1.9 Instruction formats. Arithmetic and logical class instructions shall have the ability to directly address the first 1024 words of memory and also address any location in memory by adding a displacement (up to 256 words) to one of four base registers defined in the instructions.

3.5.1.10 Double precision operations. Double precision add, subtract, load, store, and shift operations shall be provided. Multiply operations shall result in a 31-bit product. Divide operations shall provide for division of a 32-bit signed dividend.

3.5.1.11 Base and link registers. Four base and four link registers will be located with each 16,384 word section of memory. The 16 base registers will be implemented to achieve a 400 nanosecond base read cycle for those instructions which use bases in the effective address calculations.

3.5.1.12 Digital Input/Output. The Digital Input/Output Section shall provide the electrical and functional interface between the processor and the connected associated equipments. The digital I/O shall also provide three counters for internal program use as defined in 3.5.1.12.11. Space (no wiring or I/O connectors) shall be provided for a digital I/O page which can be used as future growth. The following list is a summary of the digital input/output and the intended associated equipment usage that shall be provided.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

a. INPUTS

Qty.	Type	Associated Equipment
4	50 KHz serial, asynchronous, 20 bit, 1250 words per second. Programmed interface.	Doppler Effect Fwd Looking Radar Spare Digital SINS
1	1 MHz serial, program controlled, 16 bit, 39,000 words per second.	NWDP
1	1 MHz serial, program controlled, 20 bit. 39,000 words per second.	TACAN
3	10 KHz incremental registers, asynchronous, 10,000 pulses/second/channel.	IMS
1	Special 4 bit up-down counter	Thumbwheel Encoder
3	50 KHz serial, externally controlled, 20 bit address and data. 2500 words per second/channel. DMA interface.	Spares

b. DISCRETE INPUTS

7 each, 0-28 volt logic discretes which will interface with source conditions at the Computer of:

TRUE = "1" = 28VDC = ON
FALSE = "0" = ØVDC = OFF with a source impedance greater than 20K OHMS

1 each 0-28V logic discrete which will interface with source conditions at the Computer of:

TRUE = "1" = ØVDC with a source impedance greater than 20K OHMS
FALSE = "0" = 28VDC

This discrete has a 10 millisecond filter.

81 each, 0-5V logic discretes which will interface with switches or other devices including source conditions at the Computer of:

TRUE = "1" = ØVDC = SHORT = ON
FALSE = "0" = OPEN = 5VDC = OFF with a source impedance greater than 10K OHMS

1 each, 0-5VDC logic discrete which will interface with source conditions at the Computer of:

TRUE = "1" = 5VDC = OPEN = ON
FALSE = "0" = ØVDC = SHORT = OFF

3 each, 0-5V logic discrete which will interface with source conditions at the Computer of:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

TRUE = "1" = 5VDC = ON
FALSE = "0" = ØVDC = OPEN = OFF with a source impedance greater than 10K OHMS

5 each, 0-5V logic discrete which will interface with ATE equipment with source conditions at the Computer of:

TRUE = "1" = ØVDC = SHORT = ON
FALSE = "0" = OPEN = 5VDC = OFF with a source impedance greater than 10K OHMS

5 each, internal use for TACAN parity, DIM Wait Count, TC2A identifier bit, Fail Safe, and TACAN Validity.

c. OUTPUTS

Qty.	Type	Associated Equipment
3	50 KHz serial, asynchronous, 20 bit, 2500 words per second, dual drive. DMA Interface	CH 1A, 1B Projector Map (PMDS) HUD CH 2, 2A Fwd Looking Radar, Spare, CH 3 Instrumentation, CH 3A Spare
1	1 MHz serial, program controlled, 16 bit. 39,000 words per second.	NWDP
3	200 Hz pulse from up-down counter, asynchronous	IMS
3	Serial, internally controlled, defined as follows: DMA interface.	
	1 17 bit bidirectional 1/2 duplex	CH 7A, Spare
	1 20 bit externally clocked	CH 5A, Spare
	1 50 KHz 20 bit, dual drive, 2500 words per second	CH 6A, 6B, FLIR, Spare

d. DISCRETE OUTPUTS

2 each	5 volt malfunction discretes not programmable as part of discrete output words.	IMS, ADI

11 each, 0-5V program controlled logic discretes with output characteristics at the Computer of:

True = "1" = 5 ± 1.0 VDC = OFF with a source current of 3 mA min. at 4.0 VDC
FALSE = "0" = 0.25 ± 0.25 VDC = ON with a sink current of 20 mA max.
When the Computer is off, the impedance on these outputs must be 10K OHMS or more.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

15 each, 0-5V Program controlled logic discretes and
4 each, 0-5V Program controlled logic discretes for ATE with output characteristics at the Computer of:

TRUE = "1" = 0.25 ± 0.25 VDC = ON with a sink current of 20 mA max.
FALSE = "0" = 5 ± 1.0 VDC = OFF with a source current of 3 mA min. at 4.0 VDC. When the computer is off, the impedance on these outputs must be 10K OHMS or more.

4 each, Program controlled discretes for DIM Control (3D01 to 3D04).

9 each, 0-28V Program controlled logic discretes and
1 each, 0-28V computer reliable discrete with output characteristics at the Computer of:

TRUE = "1" = 28VDC (per MIL-STD-704) = ON
FALSE = "0" = OVDC = OFF with a source impedance of 20K OHMS or more. When the Computer is off, the impedance on these outputs must be 20K OHMS or more. Each output shall be able to switch a maximum of 0.5 amperes resistive load.

1 each 0-28V discrete (2D011R) output characteristics at the Computer of:

TRUE = "1" = 28VDC (per MIL-STD-704)=ON when Computer input power is off.
NOTE: In addition to the above, 2 (1 programmable, 1D015) discretes are utilized to set the malfunction indicators on the Computer.

e. INTERRUPTS

Qty.	Type	Associated Equipment
3	Externally controlled, 1 level of priority	2 spare (populated) 1 NWDP
12	Internally controlled, 2 levels of priority	

3.5.1.12.1 50 KHz serial, asynchronous, 20 bit input channels. Each channel shall consist of a data ready control line, an address line, a data line, and a continuous 50-KHz synchronizing clock pulse line for shifting the external device register. These channels are multiplexed in a computer I/O buffer, whereby only one channel can be read at a time.

3.5.1.12.1.1 50 KHz serial, asynchronous, 20-bit input channels: data and address word format. The 20-bit data word format shall consist of 16 data bits and a control, validity, and parity bit. The 20-bit address word format shall contain seven identity bits, a control bit, and a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

parity bit. Thus for each channel, data from up to 128 devices can be separately identified. Logical zeros shall fill unused bit positions. Odd parity shall be used for the 20-bit words.

3.5.1.12.1.2 50 KHz serial, asynchronous, 20-bit input channels: channel operation. Each external device, connected to one of the serial data channels, transmits data and address words at a 50 KHz bit rate asynchronous to the program whether the Computer is receiving the data or not. A data ready pulse is transmitted coincident with the first bit of each word. When the program needs data from a serial data channel, it enables that channel by issuing a multiplexer latched control bit via the IOC Write instruction. Coincidence of data ready and the latched control bit initiates data and address word transfer into the Computer's input buffer with the least significant bit first. After one word is transferred (400 microseconds), a program interrupt is generated. Upon servicing this interrupt, the program reads and processes the addressed data from the input buffers. Reading each buffer is a 3.5 microsecond IOC instruction. Data words are sent continuously to the Computer at a 50 KHz bit rate, which corresponds to a rate of 2500 words/second. Because of the buffer reading time of the Computer, each second word is missed, reducing the reading rate to 1250 words/second. The data ready and control bits on the data and address lines shall be present for at least 16 usec in coincidence with the multiplexer control bit, in order to assure that false processing will not occur should the channel be enabled in the middle of a control bit.

3.5.1.12.2 Display update serial output. A Direct Memory Access feature which permits continuous data outputs at a 50 KHz bit rate on three channels with minimum effect on program execution time shall be provided. All three of these channels shall have parallel drive capability for feeding two external devices. Each channel shall consist of a data line, address line, a data ready line, and continuous 50 KHz synchronizing clock pulse line.

3.5.1.12.2.1 Display update serial output: data and address word format. Data and address word lengths shall be 20 bits. The data word shall contain 16 data bits and a control, validity, and parity bit. The address word shall consist of seven address bits, a control bit, and a parity bit. All unspecified bit positions shall contain logical zeros. Up to 128 different data words can be identified by the address for each channel.

3.5.1.12.2.2 Display update serial output: channel operation. When a serial data output channel has completed serial transfer of a data word to its external device, the output buffer issues a steal request to the CPU. When the instruction being executed is completed, the normal

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

program sequence is interrupted for 5.2 microseconds. During this interval, a 32-bit word containing data and address information is transferred to the output buffer. The buffer then shifts the new data and address out at a 50 KHz bit rate. The three channels are staggered in time so that the cycle steal requests occur at least 120 microseconds apart. Data transfer with three channels operating occurs at a rate of 2500 words/second/channel. No program instruction is used for these channels.

3.5.1.12.3 Serial 1 MHz input/output (NWDP). One serial 1 MHz input channel and one serial 1 MHz output channel shall be provided which will transfer data between the accumulator and an external device, under program control. This data transfers through a buffer register, which is a 16-bit serial shift register, and which is shifted under computer control.

3.5.1.12.3.1 Serial 1 MHz input/output: channel operation (NWDP). IOC Read and Write instructions requiring 19.5 microseconds are used to execute the transfers. The least significant bit is transferred first. The data word shall be identified with a 4-bit address code available on four parallel lines. Up to 16 different data words in and 16 out can be identified by the address code. Direction of data transfer shall be established by two command lines (read and write). The address lines shall be valid, while the command pulse is present, from a minimum of 1.2 microseconds before data transfer to 1.5 microseconds following data transfer. A transfer rate of approximately 39,000 words/second is possible with this channel.

3.5.1.12.4 Special 1 MHz serial input. A special 1 MHz 20-bit serial data channel shall be provided which will transfer data from an external device to the computer under program control.

3.5.1.12.4.1 Special 1 MHz serial input: data word format. The 20-bit word shall contain 16 data bits (including the sign bit), an odd parity bit and 3 bits that are not used in the TACAN interface and can be in either state. The 16 data bits shall use positions 3 though 17, with the sign bit in position 18, and the parity bit (which is parity for all 20 bits) in position 20.

3.5.1.12.4.2 Special 1 MHz serial input: channel operation. The data transmittal is initiated by program control supplying an address bit followed by a Read Command. This Read Command is accompanied by a modified bit in the OP code. The I/O will recognize the bit and force a 3.5 microsecond parallel I/O cycle. This cycle sends to the TACAN the 1 bit address of the data word the TACAN will place in its output shift

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

register. Following the 3.5 microsecond IOC, a second IOC is initiated. This command forces a 23.5 microsecond I/O which shifts the 20 bits from the external shift register to the computer.

Instruction timing is such that the first data bit will be interrogated by the TACAN parity latch a minimum of 4.5 microseconds after the TACAN is sent the first Read command. The execution of a No OP instruction between the two IOC commands will extend this minimum time to 5.5 microseconds.

3.5.1.12.5 Discrete inputs. The discrete inputs shall be combined into 7 discrete words and shall be read in units of 16 bits. The discrete input circuits shall be nonlatching. (See 3.5.1.12).

3.5.1.12.6 Discrete outputs. The discrete outputs shall be controlled by the computer by addressing one of four discrete output words. The discrete output circuits are latched outputs, except two 5 VDC and one 28 VDC. (See 3.5.1.12).

3.5.1.12.7 Parallel ATE interface. A parallel interface shall be provided through which the memory can be rapidly loaded and verified. Interface lines shall be made available for control functions and for monitoring the Storage Address Register and the Storage Buffer Register. This interface shall allow a 16-bit word to be loaded into Memory within 5 microseconds and the contents verified within another 5 microseconds.

3.5.1.12.7.1 Compatibility with CADU. The ATE interface shall be compatible with the Control and Display Unit (IBM 624983) with minimal modifications to the Control and Display Unit. If modifications are required, they shall be such that the CADU can be easily returned to its original configuration, thereby retaining its compatibility with testing of the production CP-952/ASN-91(V) Computers.

3.5.1.12.7.2 Compatibility with MLV. The ATE interface shall be compatible with the Memory Loader Verifier (MLV) (Navy Type Designation 4920-221-9637GA) with minimal modification to the MLV for extended address capability up to 64K 17 (16 bits plus main memory parity) bits. Modifications shall not cause the MLV to be incompatible with the CP-952/ASN-91(V).

3.5.1.12.8 Incremental input counters. Three 10-bit up/down counters shall be provided that can be concurrently cycled. Associated with each counter are a positive and a negative sense buffer. The input to each

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

buffer shall be a pulse on either of two counter input lines. A pulse on the positive sense line shall cause the counter to count up one. A pulse on the negative sense line shall cause the counter to count down one. The maximum acceptable combined input rate to a single register shall be 10,000 pulses/second. Pulses shall not appear simultaneously on both inputs to a given buffer. The maximum value that may be accumulated in any counter shall be sign plus 9 bits in two's complement form. Each counter shall be individually addressable. Readout and update shall be so synchronized that readout occurs without restriction. At the end of the Input/Output Control Read-Counter Instruction, the content of the counter is transferred to bit positions 6 to 15 of the accumulator and the counter cleared. These counters are intended as incremental velocity counters for stable platform accelerometer inputs.

3.5.1.12.9 Up-down counter output. Three program-loadable 8-bit up-or-down counters shall be provided to generate incremental pulses. The sign bit of each counter is sampled and buffered by the COUNT pulse, and the counter is advanced by the COUNT pulse. If the sign bit is a "0" when sampled, an up-level 5 millisecond pulse is issued on the corresponding command pulse line. If the sign bit is a "1" when sampled, a ground level 5 millisecond pulse shall be issued on the corresponding command pulse line. Each counter shall be individually addressable to load 8 bits and to read the two low-order bits. A sample pulse with a 1 microsecond ground-level pulse shall be issued on the sample line 2.56 milliseconds after the command line is valid.

3.5.1.12.9.1 Up-down counter output: operation. The two's complement negative number in the counter causes the counter to increment and a "negative" ground-level pulse to be issued on the output line until the counter reaches zero. When the counter reaches zero, an up-level pulse is issued for 5 milliseconds, and the counter is decremented. The counter continues to oscillate about zero count and to cause alternate up-level and down-level 5 millisecond pulses to be issued until the counter is reloaded. A positive number loaded in the counter causes the counter to decrement and a positive torquing commnd to be issued until the oscillations about zero count occur.

3.5.1.12.10 Interrupts. The interrupt system shall permit changing the program sequence in response to conditions external to the program. Level zero (0) shall have 6 interrupts; 3 externally controlled, 1 internal for the 50 KHz serial data input channel, and 2 internal for Signal Converter operations. Level one (1) shall have 9 internal interrupts; 3 for the interrupt counters, 1 for storage protect violation, 1 for memory parity, and 4 for DMA input channel errors. The channel errors have a supplementary mask in the least significant bit (LSB) of the Level 1 Register.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

3.5.1.12.11 Counters. Three interrupt-causing counters shall be included in the computer. The programmer can use these counters to keep track of real-time functions while continuing with other computer processing tasks.

3.5.1.12.11.1 16-Bit program loadable counters. The 16-bit program loadable counters shall provide program loadable timed interrupts. The counter shall increment at a 50 KHz rate and count up to 1.31 seconds. The 16 bits shall be loadable under program control. Reading the counter is not required. An overflow, resulting in an interrupt, results during the transition from all "1's" to all "0's". Thus, to obtain a timed interrupt, the complement of the desired time must be loaded into the counter. When the interrupt is generated, the counter shall be stopped and shall be restarted when a new value is loaded into the counter. Because of the lack of synchronization between the time at which the counter is loaded and the 50 KHz input, an allowable error up to minus 20 microseconds may exist between the loaded time value and the actual time to the generated interrupt. The stepping pulse is applied to the sixteenth bit.

3.5.1.12.11.2 Go-No-Go counter. A 16-bit counter shall be included for use as a Go-No-Go counter. The counter shall be resettable and readable by the program and serve as a real-time counter. The counter shall generate an interrupt if not reset within 1.31 seconds maximum. If not reset, the counter shall turn on computer fault discretes, cause I/O data wrap-around, and continue to count. The input to the interrupt register shall rise every 2.62 seconds thereafter. This counter shall be incremented at a 50 KHz rate. Because of the lack of synchronization between the time at which the counter is reset and the 50 KHz input, an allowable error up to 20 microseconds may exist between the time value read and the actual time since resetting.

3.5.1.12.12 50 KHz serial externally controlled 20-bit input channels. Each of three channels shall consist of a Data Ready control line, an Address line, a Data line, and a Clock line. On two of these channels the clock shall be a 50 KHz continuously running clock for synchronizing the shifting of data on to the Data line. On the remaining channel, the clock line is externally sourced with a switching rate up to 100 KHz. These channels operate independently and can thus all be receiving data.

3.5.1.12.12.1 50 KHz serial externally controlled 20 bit input channels: data and address word format. The 20 bit data word format shall consist of 16 data bits and a control, validity, and parity bit. The 20 bit address word format shall contain eight identity bits, a control bit, and a parity bit. Thus for each channel, data from up to 256 devices

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

can be separately identified. Logical zeros shall fill unused bit positions. Odd parity shall be used for the 20-bit words.

3.5.1.12.12.2 50 KHz serial externally controlled 20-bit input channels: channel operation. Each external device, connected to one of the serial data channels, transmits data and address words at a 50 KHz bit rate asynchronous to the program whether the Computer is receiving the data or not. A data ready pulse is transmitted coincident with the first bit of each word. When the program needs data from a serial data channel, it enables that channel by issuing a CIOS instruction to enable that channel with its control word. Coincidence of data ready and the control bit initiates data and address word transfer into the input buffer with the least significant bit first. After one word is transferred, it is moved to a buffer so the next word in line may be received. The buffered word is checked for good parity and validity, then moved to the memory location defined by the address word added to 2048. Every word is read and thus, the maximum transfer rate is 2500 words per second on each channel.

3.5.1.12.13 Serial, externally controlled output channels. Three output channels shall be provided in addition to those described in paragraph 3.5.1.12.2.

3.5.1.12.13.1 17-bit bidirectional half duplex channel. A channel shall be provided which has the capability to interface with the input/ output of the Mid-Course Guidance Computer of the Harpoon Missile. This channel shall send out data under DMA control at a bit rate of 100 KHz. When a word has been partly shifted out, a new 32-bit word containing data address information shall be transferred to a buffer from the CPU by DMA. The channel must be initiated under Program Control. This 17-bit channel has the capability of reading an occasional word. This shall be buffered in the Computer for the program to read.

3.5.1.12.13.2 50 KHz 20-bit output channel. DMA shall permit continuous data outputs at a 50 KHz bit rate with minimum effect on program execution time. This channel shall have parallel drive capability for feeding two external devices. The channel shall consist of a data line, address line, a data ready line, and continuous 50 KHz synchronizing clock pulse line.

3.5.1.12.13.2.1 50 KHz 20-bit output channel: data and address word format. Data and address word lengths shall be 20 bits. The data word shall contain 16 data bits and a control, validity, and parity bit. The address word shall consist of seven address bits, a control bit, and a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

parity bit. All unspecified bit positions shall contain logical zeros. Up to 128 different data words can be identified by the address for each channel.

3.5.1.12.13.2.2 50 KHz 20-bit output channel: channel operation. When a serial data output channel has completed serial transfer of a data word to its external device, the output buffer issues a DMA request to the CPU. When the instruction being executed is completed, the normal program sequence is interrupted for 5.2 microseconds. During this interval, a 32-bit word containing data and address information is transferred to the output buffer. The buffer then shifts the new data and address out at a 50 KHz bit rate. Data transfer occurs at a rate of 2500 words/second/channel. This channel may be initiated or turned off under Program Control.

3.5.1.12.13.3 20-bit externally clocked channel. This channel shall operate identically to the channel described in 3.5.12.13.2.2 except that the external device shall provide the data shifting clock and may operate it at any rate up to 100 KHz including pause. Data transfer rate is dependent upon clock (5000 words/second at 100 KHz).

3.5.1.12.14 Special 4-bit up-down counter. A 4 bit plus sign program readable up-down counter shall be used to accept as inputs two square waves which are displaced from each other by 90 ±30 degrees and which are generated by a thumb driven modified optical encoder. The frequency of the input may vary from 0 to 250 pps. Each pulse #1 leading pulse #2 shall cause the counter to increment, and each pulse #1 trailing pulse #2 shall cause the counter to decrement. Readout shall be the accumulator positions 11 to 15 with the complement of the sign in position 11, the most significant bit in position 12 and the least significant bit in position 15. Accumulator positions 8-10 are zero. The high order 8 bits of the accumulator contain a nonvalid bit pattern which must be ignored. The counter shall clear after readout. Attempts to prevent loss of pulses which occur during readout is not required.

3.5.1.13 Analog I/O. An all solid state analog I/O section shall be provided for A/D and D/A conversions. The following is a summary of the requirements:

a. INPUTS

7 Synchro to Digital
8 DC Voltage to Digital

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

b. OUTPUTS

4	Digital to Synchro Control Transformers
3	Digital to Synchro Torque Receivers
4	Digital to DC Voltage
1	Digital to AC Voltage

c. INPUT/OUTPUT DATA FORMAT

D/A Data Format	Synchro:	Tan θ/cot θ + octant
	DC:	Magnitude + polarity
	AC:	Magnitude + Sign (in phase)
		Magnitude - Sign (out of phase)
A/D Data Format	Synchro:	K Sin θ/K Cos θ
	Linear:	Magnitude + polarity

d. RESOLUTION

A/D:	12 bits magnitude + sign, Two's Complement
Digital/Synchro:	3 bits octant + 10 bits tan θ or cos θ, binary
Digital/DC:	10 bits + sign, Two's Complement
Digital/AC:	10 bits + sign, Two's Complement

e. ACCURACY

Synchro A/D:	±6 min. maximum
Synchro A/D Conversion Jitter Spread:	6 arc min. maximum

Spread definition: The spread for any given 200 consecutive conversions, at 25 per second min., shall be determined by subtracting the lowest reading from the highest reading and this difference shall not exceed 6 arc minutes.

Digital to Torque Receiver:	±3.6 degrees maximum
Digital to Ctrl XFMR:	±6 min. maximum
DC (Bull Pup Hand Ctrl) A/D (relative):	±0.25% maximum
DC (25V) A/D (absolute):	±1.0% maximum
DC A/D (relative):	±0.1% maximum
DC D/A (absolute):	±0.5% maximum
AC D/A (relative to AC ref.):	±0.2% maximum

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the currentversion before use.

f. CONVERSION SPEED

A/D Conversion Speed (synchro)	2.5 msec. per conversion
A/D Conversion Speed (dc)	.332 msec. per conversion
D/A Conversion Speed (synchro)	1.8 msec. per conversion
D/A Conversion Speed (dc)	50 usec. per conversion (excluding slew rate of driver
D/A Conversion Speed (synchro torque)	1.8 msec. (36 degrees change): 2 sec. (180 degrees change)
D/A Conversion Speed (AC)	1.8 msec. per conversion

g. OUTPUT LOAD DRIVING CAPABILITIES:

Digital to Synchro Control Transformers Digital to Synchro Torque Receivers	2 Control Transformers - EPAY -500-5 Each output - 2.6 amps minimum P-P drive at 320 milliseconds ±2% at 40% ±2% duty cycle.

Digital to DC Voltage:

1 - 0 to 18V	-	10K OHMS min.
2 - ±18V		1K OHMS min.
1 - ±4.0V		1000 OHMS

Digital to AC Voltage:

1 - 0 to 11.8 RMS - 10K OHMS

3.5.1.13.1 A/D and D/A conversion. Only one A/D conversion or one D/A conversion at a time shall be required. The capability to perform an A/D conversion simultaneously with a D/A conversion shall not be required.

3.5.1.13.2 A/D conversion. An IOC Read command will start the analog to digital conversion. When the conversion is complete, a Data Ready signal will cause program interrupt. When the interrupt is serviced, another IOC Read Command is used to transfer the data from the data register to the accumulator. The A/D converter is reset only with an IOC Read Command with all ones address. For a synchro conversion, two Read commands are required: one for the sine and one for the cosine.

3.5.1.13.3 D/A conversion. An IOC Write Command will start the digital to analog conversion. The converter will convert the digital word to the analog signal and at the end of this process will issue a converter not busy interrupt.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:48Z
Check the source to verify that this is the current version before use.

3.5.1.14 Fail-Safe. All critical output signals from the computer shall be inhibited as a result of any fail mode, warmup, shutdown, turn off, or as a result of abnormal power as defined in 3.3.12.

3.5.1.14.1 Fail-Safe signals. The following signals shall be fail-safed:

a. Fire Ready, 1D010

b. Bomb Release, 1D011

c. Cursor Slave, 1D013R

d. Antenna Slave, 1D014R

e. Nonalign, 2D010R

f. Computer Control, 2D014

g. All Populated Spare Outputs except 1D0 5, 6, 7, 8, and 2D016

h. All serial channels shall be fail safed by inhibiting the Data Ready control pulse except CH3 and CH3A.

i. All computer status indicator signals to connected equipment shall be set to the fault condition.

j. Discrete input word 1, bit 16 shall be set to a logical "1".

k. Analog I/O outputs shall be in an indeterminate state for a period of ten milliseconds in any fail mode shutdown, turn off, or as a result of abnormal input power as defined in 3.3.12. The Read/Write commands to the Analog I/O shall be inhibited during any of these conditions. At the end of this period, the DC outputs shall appear as zero (0) volts with the source impedance of 50 ohms maximum and the D/S and wheels output channels will go to zero (0) degrees.

3.5.1.15 Power supply. All DC voltages and sequencing required to operate the Computer and the NWDP shall be contained in the Computer. The primary power source shall be as defined in 3.3.12.

3.5.2 NWDP C-7830A/ASN-91(V), C-7831/ASN-91(V).

3.5.2.1 Function. The NWDP shall display positional and guidance information and convey operator inserted information and control to the associated equipments via the Computer.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

3.5.2.2 Form factor. The size and shape of the NWDP shall be as shown in IBM Drawing 6877727.

3.5.2.3 Weight. The weight of the NWDP shall not exceed 7.0 pounds.

3.5.2.4 Controls. The general arrangement of the controls and displays shall be in accordance with Figure 1. The controls and displays of C-7830A/ASN-91(V) are red-edge lighted in accordance with IBM 6870210. The controls and the displays of the C-7831/ASN-91(V) are white-edge lighted in accordance with IBM 6870299. The front panel shall contain the following controls and displays.

a. Computer Power Switch - A three position toggle switch will control power to the computer. The PWR position applies +28 Volts DC (Aircraft) supply. The Test position is a momentary position which sends a +5 VDC logic signal to the Computer to initiate self-test. This toggle switch is a pull-to-unlock type in the OFF and PWR positions.

b. Auxiliary Equipment Power Switch - A three position toggle switch will control power to Auxiliary Equipment. The "Y" position can switch +28 VDC (Aircraft), at 3 AMPS max, to Auxiliary Equipment. The "Z" position can send a ±5 VDC logic signal to Auxiliary Equipment for control, etc. The "X" position is open.

c. "A" Thumbwheel Switch - A four position BCD type internally illuminated thumbwheel switch will provide means for entering data into the Computer. Nomenclature will be 1, 2, 3, and 4. Illumination will be controllable from the aircraft master dimmer.

d. "B" Thumbwheel Switch - A five position BCD type internally illuminated thumbwheel switch will provide means for entering data into the Computer. Nomenclature will be 1, 2, 3, 4, and 5. Illumination will be controllable from the aircraft master dimmer.

e. "C" Thumbwheel Switch - A three position BCD type internally illuminated thumbwheel switch will provide means for entering data into the Computer. Nomenclature will be 1, 2, 3. Illumination will be controllable from the aircraft master dimmer.

f. "D" Thumbwheel Switch - A six position BCD type internally illuminated thumbwheel switch will provide means of entering data into the computer. Nomenclature will be 1, 2, 3, 4, 5, and 6. Illumination will be controllable from the aircraft master dimmer.

g. Fly-to Thumbwheel Switch - A ten position BCD type internally illuminated thumbwheel switch will enable selected Destination or Mark coordinates to be used in computer flight calculations. Nomenclature

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the currentversion before use.

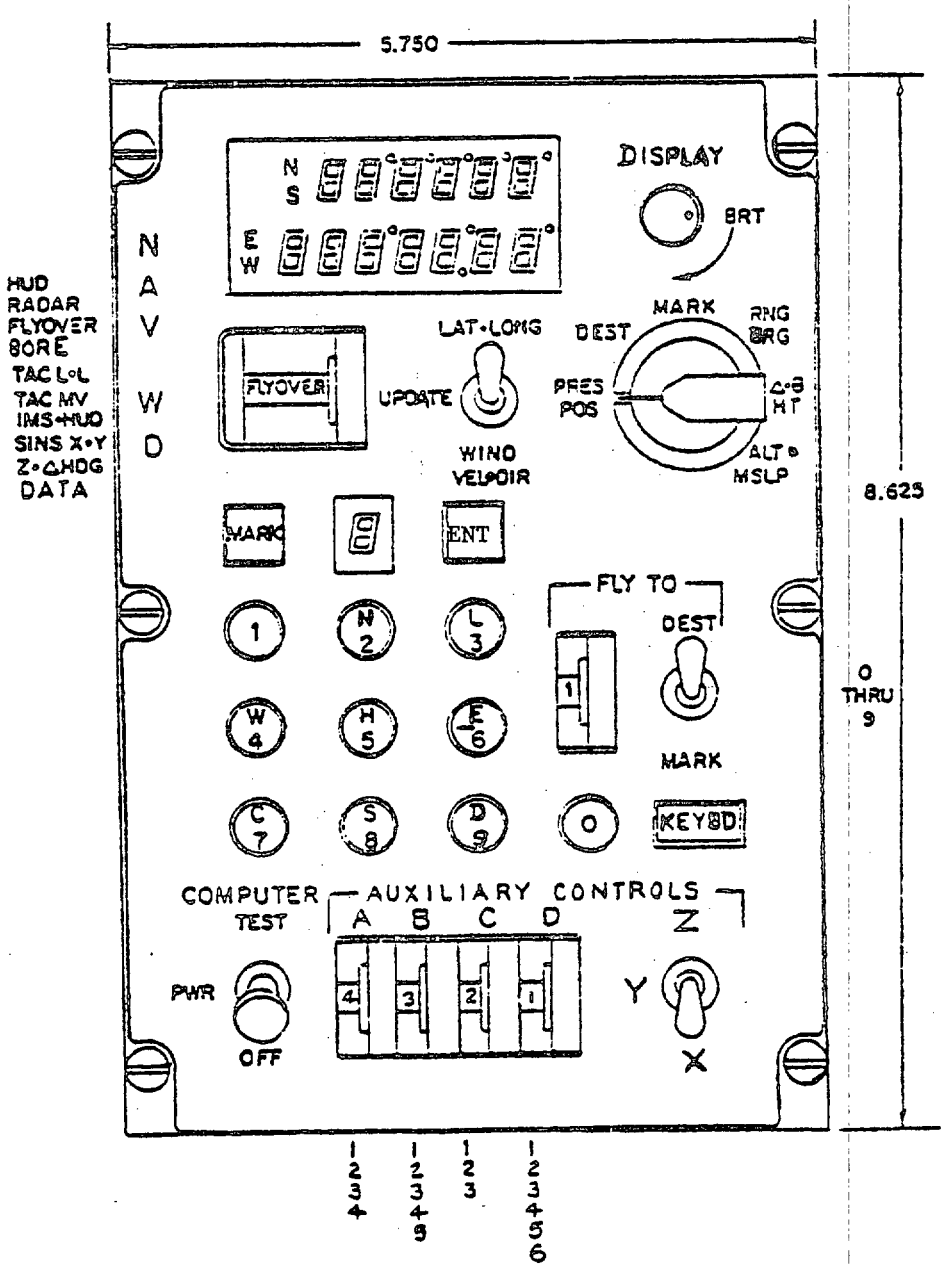


FIGURE 1. Front panel layout - navigation/weapon delivery panel (NWDP).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

will be 0 through 9. The switch is used in conjunction with the Fly-To toggle switch. Illumination will be controllable from the aircraft master dimmer.

h. Fly-to-Toggle Switch - A two position toggle switch will enable the pilot to select either a destination or mark to be used in computer flight calculations. Nomenclature will be DEST and MARK. The switch will be used in conjunction with the Fly-To thumbwheel switch.

i. Keyboard Pushbutton Switches - Ten momentary contact SPST low-level illuminated pushbutton switches will enable data to be inserted into the Computer and recalled for display from the Computer. The lenses will be opaque except for the nomenclature which will be diffused white backed by a red or white filter. Nomenclature will be:

N	L	W	H	E	C	S	D
1,	2,	3,	4,	5,	6,	7,	8,	9,	0

Illumination will be controllable from the aircraft master dimmer.

j. Keyboard Enable Pushbutton Switch - A momentary contact SPST pushbutton switch will enable the Keyboard switches to be used for inserting data into the Computer. The lens will be low-level lit when unselected. When selected, the nomenclature KEYBD will be lit on an opaque background at a higher level of illumination. The lens will be opaque except for nomenclature.

k. Enter Pushbutton Switch - A momentary contact SPST pushbutton switch will enable the Computer to accept data inserted via the keyboard switches when KEYBD switch is high-level illuminated, and will enable the Computer to update present position when the KEYBD switch is low-level illuminated. Lens will be opaque when unselected. When selected, the lens will be opaque except for the nomenclature ENT which will be diffused white backed by a green filter.

l. Mark Pushbutton Switch - A momentary contact SPST low-level illuminated pushbutton switch will enable the pilot to insert the present position coordinates into computer storage for future reference. The lens will be opaque except for the nomenclature MARK which will be diffused white backed by a red or white filter. Illumination will be controllable from aircraft master dimmer.

m. Mode Rotary Switch - A six position rotary switch (with growth capability to eight positions) will enable the pilot to select the operational mode of the Computer System. Control is provided over the items to be inserted into the Computer and displayed on the NWDP. This switch is used in conjunction with the PRESENT POSITION toggle switch and UPDATE thumbwheel switch. Nomenclature around the switch will be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

PRES POS, DEST, MARK, RNG BRG, Δ·BHT, ALT·MSLP. The knob will be gray colored, with a backlighted illumination line.

n. Present Position Toggle Switch - A three position toggle switch will enable the further subdivision of PRESENT POSITION into LAT·LONG, UPDATE, or WIND VEL·DIR. This switch is used in conjunction with the MODE rotary switch and UPDATE thumbwheel switch.

o. Update Thumbwheel Switch - A ten position BCD type internally illuminated thumbwheel switch will enable mode selection and data to be inserted into the Computer on the carrier deck and in flight. This switch is used in conjunction with the MODE rotary switch and PRESENT POSITION toggle switch. Nomenclature will be HUD, RADAR, FLYOVER, BORE, TAC L·L, TAC MV, IMS·HUD, SINS X·Y, Z·ΔHDG, DATA. Illumination will be controllable from the aircraft master dimmer.

p. Mark Numeric Indicator - An incandescent seven segment numeric indicator will enable a display of MARK selected to be present on the NWDP. The number displayed will represent the MARK just selected by depressing the MARK switch. A Polaroid filter will be placed on the indicator for enhanced visibility in high ambient light. An external (users system) generated signal will light all lamps for lamp test.

q. Position/Navigation Display - Thirteen incandescent seven segment indicators and two incandescent two-character indicators, will enable position and navigation data to be displayed. A Polaroid filter will be placed on the display for enhanced visibility in high ambient light. An aircraft-generated signal will light all lamps for lamp test.

r. Dimmer Control - A potentiometer will be used to adjust the light intensity of the POSITION/NAVIGATION DISPLAY and MARK NUMERIC INDICATOR for various ambient lighting conditions. Clockwise rotation of the potentiometer will increase light intensity. The knob will be gray colored.

3.5.2.5 Operation. Data shall be transferred between the NWDP and the Computer via a 16 bit 1 MHz program controlled serial channel. Direction of data flow shall be controlled by Read (output from panel), Write (input to panel) control lines. Data identity shall be controlled by 4 address lines. Data shall be shifted in or out of the panel by shift and clock signal from the Computer. See 3.5.1.12.3.1 for Channel operation.

3.5.3 Base, shock mount, electrical equipment, MT-4078/ASN-91(V). A shock mounting shall be provided for the Computer. The mounting shall use self-locking clamps.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

3.5.3.1 Form factor. The size and configuration shall be in accordance with IBM Drawing 6877502.

3.5.3.2 Weight. The weight of the Shock Mount shall not exceed 8.4 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests. The Computer covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements. The NWDP shall be subjected to tests in accordance with IBM Procedure 76-329-2410.

a. Preproduction (First Article) Tests.

b. Acceptance Tests.

c. Life Tests.

4.2 Preproduction tests. Preproduction tests when specified in the contract shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The cognizant Navy office shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government. Preproduction testing shall not be performed on the NWDP.

4.2.1 Scope of tests. Preproduction tests and test methods shall be determined by the Contractor and shall include tests to determine that the system will satisfactorily meet the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

Preproduction tests for the Computer shall be run in accordance with MIL-T-5422 and shall be limited to the following tests, test limitations, and applicable specification requirements in Section 3 herein.

a. Vibration - Curve I of Figure 5 MIL-E-5400 extended to 2000 Hz at the ±10g level for vibration.

b. Temp. Alt. - In accordance with MIL-T-5422, Paragraphs 3.3.10.2 and 3.3.10.3 herein applies.

c. Shock - In accordance with MIL-T-5422.

d. EMI - Paragraph 3.3.6 herein applies.

The following environmental requirements shall be verified by similarity data or analytical results.

a. Humidity

b. Salt Fog

c. Explosion

d. Sand and Dust

e. Fungus

4.2.2 Preproduction approval. Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to approval of the preproduction sample. Fabrication of production equipment prior to approval of the preproduction sample is at the contractor's own risk.

4.2.3 Preproduction rejection and retest. If the sample item fails to pass any preproduction test requirement, full particulars of the failure, corrective action taken, and subsequent test results shall be reported at the time the action is taken by the Contractor to the cognizant Navy Office for approval. Product drawings, schematics, etc., shall reflect all agreed upon changes prior to production and delivery of the equipment.

4.3 Initial production tests. Initial production tests shall not be required as part of this specification.

4.4 Acceptance tests. The Contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the cognizant Navy

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

Office. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the Government. The Contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the Contractor or laboratory, as applicable.

Acceptance of approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests

b. Sampling Tests (Not Applicable)

c. Reliability Assurance Tests

d. Special Tests (Not Applicable)

4.4.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests in accordance with IBM Procedure 76-329-2440. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product

b. Operational Test

c. Manufacturing Run-In Test

4.4.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing run-in test. Each unit shall receive a minimum of 40 cycles (200 hours burn-in) with the last 10 cycles (50 hours) failure free. The temperature profile and power cycling shall be in accordance with IBM Procedure 76-329-2408. Vibration periods shall be required. These tests shall be followed by an operational test prior to shipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

4.4.1.3.1 Countable pre-burn-in time. Each unit in accordance with IBM Prodedure 76-329-2409, shall have successfully completed the 150-hour Pre-Burn-In Test when it meets the following requirements:

a. The Computer shall have 150 hours of countable burn-in time.

b. All SRA's shall have a minimum of 100 hours of countable burn-in time.

c. Not more than three SRA's in total shall have less than 150 hours of countable burn-in time out of which:

(1) Not more than one Memory, CPU or I/O SRA shall have less than 150 hours of countable burn-in time.

(2) Not more than one SDR in the power supply shall have less than 150 hours of countable burn-in time.

(3) Not more than one signal converter SRA shall have less than 150 hours of countable burn-in time.

d. The computer must be free of all solid failures (including repeatable temperature sensitive ones (i.e., cold-start failures).

Each unit must meet this requirement before accountable time can begin on the 50-hour failure-free burn-in.

4.4.1.3.2 Countable burn-in time - 50 hours failure free. Each serialized subassembly shall have accumulated a minimum of 50 hours burn-in, failure free. Time for a failed subassembly shall reset to 150 hours. A failed subassembly in a unit undergoing 50 hours failure-free test may be replaced with a subassembly having some minimum number of failure-free hours. Failure-free tests may be resumed on the unit with the countable time set equal to the hours on the subassembly having the least number of countable failure-free hours.

4.4.1.3.3 Subsequent test requirements. Pre-burn-in test requirements for SRA's used for replacement in units failing the 50-hour burn-in or subsequent FAT or CAT tests shall not be imposed.1/ Test procedures, IBM No. 76-329-2440 and 76-329-2410 contain the requirements for FAT and CAT.

1/Exception: As the majority of units shipped will exceed the requirements of 4.4.1.3.1, replacement SRA's intended for use in Rel Demo units shall have a minimum of 100 hours of pre-burn-in and/or burn-in time in order to realistically represent the deliverable units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

4.4.1.3.4 Spares test requirements. Unit (new build) being delivered as spares shall be tested in accordance with the requirements for production units. SRA's (new build) being delivered as spares shall have accumulated 25 hours failure free countable burn-in time prior to CAT.

4.4.1.3.5 Recycles. Recycle units requiring "new build" replacement subassemblies shall require replacement subassemblies which meet the spares test requirements only.

4.4.1.3.6 Test incidents and failure reporting. All test incidents occurring after start of burn-in shall be investigated, failures analyzed, appropriate corrective action determined, and the following data provided:

a. Failure Reports

b. Failure Analysis Reports

c. Failure Analysis and Corrective Action Summaries

d. Reliability Test Reports

Corrective actions shall be implemented subject to approval by the cognizant Navy office.

4.4.2 Sampling tests. Sampling tests shall not be required as part of this specification.

4.4.3 Reliability assurance tests. Tests as required by both the Qualification Phase (when required by contract or order) and the All Equipment Tests shall be conducted.

4.4.3.1 Qualification phase. When required under the contract or order Reliability Assurance Qualification Phase Tests shall be performed in accordance with MIL-STD-781. The test plan, test level, number requirements, MTBF and accept/reject criteria including the percent of risk shall be determined at the time such tests become part of the contract or order.

4.4.3.2 All equipment test. Each computer produced except those submitted for Qualification Phase Tests shall be tested in accordance with the Manufacturing Run-In Tests paragraph 4.4.1.3 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

4.4.4 Special tests. Special tests shall not be required as part of this specification.

4.5 Life tests. Life tests in the form of TAAF Reliability Growth Tests shall be conducted on two (2) computer units per IBM Procedure 76-329-2413. Two (2) computers and two (2) NWDP shall be furnished as GFE for these tests. The duration of test shall be 2000 hours of total sample test time (cumulative test duration). The NWDP are required for control of the operation of the computers. The NWDP shall be operated outside the environmental chamber. Failures of the NWDP shall therefore be excluded from the requirements of these tests. The computers shall be mounted in the production Burn-in base shock mounts when conducting these tests.

4.5.1 Environmental profile. The sample(s) under test shall be subjected to the Burn-in Profile of IBM Procedure 76-329-2413. The vibration, temperature cycling and input power variations while functionally operating shall be to the following requirements.

a. Vibration: 2.2g's peak, 25 ±5 Hz 10 minutes period

b. Temperature Cycling: -65°F cold soak to +160°F ambient

c. Voltage Variation:

115 VAC 400 Hz	High	121.0 ±3.5 VAC
	Low	105.0 ±3.0 VAC
28 VDC	High	30.0 ±1.0 VDC
	Low	24.0 ±1.0 VDC

4.5.2 Performance check. Testing shall be performed in a Self Test Mode with test stand power and the NWDP for control. Sufficient characteristics of the Operational Tests shall be checked to assure satisfactory equipment operation.

4.6 Test procedures. The procedures used for conducting preproduction tests, acceptance tests, and life tests shall be prepared by the Contractor and submitted to the cognizant Navy office or its designated representative for review and approval. The right is reserved by the cognizant Navy office or its designated representative to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

will be provided and may be used when their use is approved by the cognizant Navy office or its designated representative. However, the right is reserved by the cognizant Navy office or its designated representative to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Presubmission testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and retest. Equipment which has been rejected, may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and action taken to correct the defects found in the original shall be furnished to the cognizant Navy office.

5. PREPARATION FOR DELIVERY.

5.1 General. Unless otherwise specified in the contract or order all major units and parts of the equipment shall be preserved, packaged, packed and marked for Level A in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794 and Method II A of MIL-P-116.

6. NOTES

6.1 Intended use. The original intended use of the system procured to this specification is to be used in the weapon release system armament and navigation system circuitry of the A-7 high performance, light attack aircraft.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.

a. Title, number, and date of this specification

b. Selection of applicable levels of packaging and packing if different than that specified in 5.1.

6.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any specification.

b. This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the cognizant Navy office.

c. Referenced Specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type designations. The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Associated equipment

Equipment	Type Designation	Spec. No.
Altimeter, Radar	AN/APN-141 or AN/APN-194	MIL-A-23887A MIL-A-81605B
Computer, Air Data	CP-953/AJQ	204-16-35 1/
Horizontal Situation Indicator	ID-1013/A or AQU-6/A	MIL-I-23366A1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

Radar Doppler	AN/APN-190(V)	204-16-181/
System, Forward Looking Radar	AN/APQ-126	204-16-131/
Attitude-Director Indicator	ID-1329/A or	MIL-I-23524-3
	ARU-21/A	MIL-I-27619E
Set, TACAN	AN/ARN-84	MIL-N-81207
Unit, Armament Station Control	C-8185/AWE	DIR-2-53500/218-511 1/
	C-8230/AWE	DIR-2-53500/216-101 1/
	C-8652/AWE or	DIR-2-53500/216-101 1/
	C-9387/AWE	204-16-302 1/
Set, Pilot, Head-Up Display	AN/AVQ-7(V)	204-16-19 1/
		204-16-282 1/
Switch, Pilot's Stick Grip		
Set, Inertial Measurement	AN/ASN-90(V)	204-16-70 1/
Projector Map Display Set	AN/ASN-99 or	204-18-14 1/
	AN/ASN-99A	
Angle of Attack Transducer		204-15-361 1/
Coupler Converter	CU 1923/AS	204-18-44B1/
Armament Release Control Panel (hereinafter referred to as AR Panel)		

1/ The issue in effect on date of contract go-ahead to the computer vendor.

6.7 Nonrepairable subassemblies. As a general rule, nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts, tolerances, and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the nonrepairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications, this figure must be nearer to 50,000 hours.

6.8 Definition of terms.

6.8.1 Unit. A removable box, made of modules, that connects with other units by external cabling and connectors, as typified by a gyro platform unit, a horizontal display unit, an interface unit, etc.

6.8.2 Modules. A subdivision of a unit characterized by the ease with which it can be removed by the application of a screwdriver to hold-down screws or clamps.

6.8.3 Card. An item removable by mechanical disconnect and made up of microelectronic integrated circuits (MIC), components, and elements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

6.8.4 Microelectronic integrated circuits (MIC). An integral silicon planar epitaxial structure that performs functions such as those of gates, amplifiers, adders, bi-stable elements, etc.

6.8.5 WRA. Weapons Line Replacement assembly; an assembly which for maintenance purposes is replaceable at the flight line or flight deck.

6.8.6 SRA. Shop Replaceable Assembly - a module for maintenance purposes is replaceable in the line shop or intermediate level of maintenance.

6.8.7 TAAF. Test, analyze, and fix.

6.8.8 Failure. Cessation of equipment operation or degradation of a function beyond specified limits. Failure shall be considered either relevant or nonrelevant in accordance with the following criteria:

a. Nonrelevant failures as defined below shall not be classified as failures.

(1) Malfunctions caused by the operation of the equipment outside the specified electrical, mechanical, or temperature limits.

(2) Malfunctions caused by the operation or malfunction of equipment(s) external to the WRA.

(3) Malfunctions which are humanly induced through negligence, accidental overstress, improper connection, damage in handling, installation or operation.

(4) Apparent malfunctions which result from programming errors.

(5) Malfunctions that have exhibited intermittent symptoms which cannot be verified or cannot be confirmed by reoccurrence within 24 hours of fault-free operation.

b. All failures not covered in (a) above shall be considered relevant.

6.8.9 CAT - Customer Acceptance Test.

6.8.10 FAT - Factory Acceptance Test.

6.9 Cognizance. This specification is under the cognizance of AIR-533.

Preparing Activity
(NAVY-AS)

Project No. FSC-5826-N145

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

TABLE I.

COMPUTER INSTRUCTION SET

0-Register 0 1 2 3 4 5 6 7	Instructions	Mnemonic	Time (Microseconds) Direct	Relative
0 0 0 0 0 0 0 0	NO OPERATIONS	NOP	1.0	
0 0 0 0 0 0 0 1	ABSOLUTE VALUE OF ACCUMULATOR	AVA	1.0	
0 0 0 0 0 0 1 0	COMPLEMENT LOGICAL	CL	1.0	
0 0 0 0 0 0 1 1	COMPLEMENT ARITHMETIC	CA	1.0	
0 0 0 0 0 1 0 0	ZERO A & Q	ZAQ	1.0	
0 0 0 0 0 1 0 1	ZERO Q REGISTER	ZQ	1.0	
0 0 0 0 0 1 1 0	EXCHANGE A & Q	XAQ	1.0	
0 0 0 0 0 1 1 1	SAVE ACCUMULATOR & ZERO	SAZ	1.0	
0 0 0 0 1	DIAGNOSE	DIAGNOSE	1.8	
0 0 0 1 0	COMPARE BYTE IMMEDIATE & SKIP	CBIS	1.4	
0 0 0 1 1 0 0 1	BRANCH INDIRECT	BIA	2.0	
0 0 0 1 1 0 1 0	LOAD STATUS WORD	LSW	1.8	
0 0 0 1 1 1 0 0	STORE STATUS WORD	STSW	1.8	
0 0 0 1 1 1 1 X	LOAD BASE RELATIVE	LBR	3.8	
	BRANCH FORWARD ON CONDITION			
0 0 1 0 0 0 0 0	CARRY	BFCY	1.0/1.2	
0 0 1 0 0 0 0 1	PLUS	BFP	1.0/1.2	
0 0 1 0 0 0 1 0	ZERO	BFZ	1.0/1.2	
0 0 1 0 0 0 1 1	NOT MINUS	BFNM	1.0/1.2	
0 0 1 0 0 1 0 0	MINUS	BFM	1.0/1.2	
0 0 1 0 0 1 0 1	NOT ZERO	BFNZ	1.0/1.2	
0 0 1 0 0 1 1 0	NOT PLUS	BFNP	1.0/1.2	
0 0 1 0 0 1 1 1	UNCONDITIONAL	BF	1.0/1.2	
0 0 1 0 1	BRANCH	B	1.4	
	BRANCH BACKWARD ON CONDITION			
0 0 1 1 0 0 0 0	CARRY	BBCY	1.0/1.2	
0 0 1 1 0 0 0 1	PLUS	BBP	1.0/1.2	
0 0 1 1 0 0 1 0	ZERO	BBZ	1.0/1.2	
0 0 1 1 0 0 1 1	NOT MINUS	BBNM	1.0/1.2	
0 0 1 1 0 1 0 0	MINUS	BBM	1.0/1.2	
0 0 1 1 0 1 0 1	NOT ZERO	BBNZ	1.0/1.2	
0 0 1 1 0 1 1 0	NOT PLUS	BBNP	1.0/1.2	
0 0 1 1 0 1 1 1	UNCONDITIONAL	BB	1.0/1.2	
0 0 1 1 1	INPUT/OUTPUT CONTROL	IOC	3.5/19.5/23.5	
0 1 0 0 0	ADD DOUBLE	AD	2.8	3.2
0 1 0 0 1	ADD	A	1.8	2.2
0 1 0 1 0	STORE DOUBLE	STD	2.8	3.2
0 1 0 1 1	STORE ACCUMULATOR	STA	1.8	2.2
0 1 1 0 0	SUBTRACT DOUBLE	SD	2.8	3.2
0 1 1 0 1	SUBTRACT	S	1.8	2.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

TABLE I.(cont'd)

COMPUTER INSTRUCTION SET.

0-Register 0 1 2 3 4 5 6 7	Instructions	Mnemonic	Time (Microseconds) Direct	Relative
0 1 1 1 0	LOAD DOUBLE	LD	2.8	3.2
0 1 1 1 1	LOAD ACCUMULATOR	LA	1.8	2.2
1 0 0 0 0	SHIFT LEFT AND NORMALIZE	SLN	2.8 1/	
1 0 0 0 1	DIVIDE	D	8.2	8.6
1 0 0 1 0	MODIFY REGISTER (DISP ≠ 0)	MDR	3.0	
1 0 0 1 0	STORE INSTRUCTION COUNTER (DISP = 0)	STIC	1.8	
1 0 0 1 1	CHANNEL INPUT OUTPUT SERVICE	CIOS	2.0	
1 0 1 0 0	SET BIT TO MEMORY	SBM	3.0	4.0
1 0 1 0 1	MULTIPLY	M	7.8	8.2
1 0 1 1 0	LOAD REGISTER	LR	2.6	
1 0 1 1 1	EXCLUSIVE OR	X	1.8	2.2
1 1 0 0 0	SHIFT RIGHT DOUBLE & ROTATE (D = 000xxxxx)	SRDR	1.6 1/	2.0 1/
1 1 0 0 0	SHIFT RIGHT DOUBLE ARITHMETIC (D = 010xxxxx)	SRDA	1.6 1/	2.0 1/
1 1 0 0 0	SHIFT RIGHT DOUBLE LOGICAL (D = 011xxxxx)	SRDL	1.6 1/	2.0 1/
1 1 0 0 0	SHIFT LEFT DOUBLE & ROTATE (D = 100xxxxx)	SLDR	1.6 1/	2.0 1/
1 1 0 0 0	SHIFT LEFT DOUBLE (D = 110xxxxx)	SLD	1.6 1/	2.0 1/
1 1 0 0 1	COMPARE	C	2.8	3.2
1 1 0 1 0	BRANCH FORWARD ACCUMULATOR BIT	BAB	1.4	
1 1 0 1 1	AND	N	1.8	2.2
1 1 1 0 0	STORE REGISTER	STR	2.2	
1 1 1 0 1	TALLY	T	3.0	3.4
1 1 1 1 0	BRANCH INDIRECT AND LINK	BIAL	2.8	
1 1 1 1 1	OR	O	1.8	2.2

1/ = +0.2n (n = the number of positions to be shifted)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:48Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-85016(AS) Computer Set, Tactical AN/ASN-91B(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:48Z
Check the source to verify that this is the current version before use.